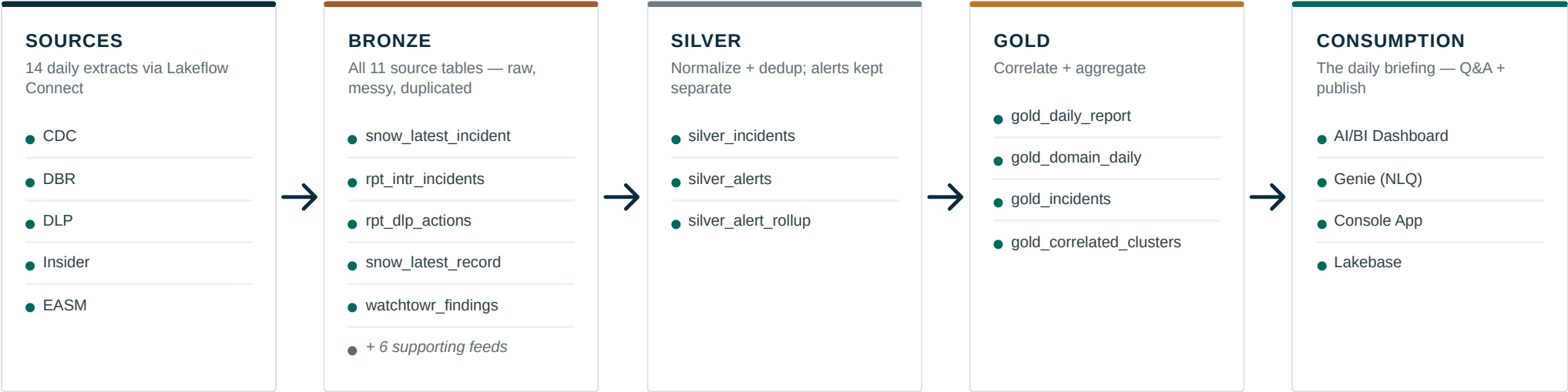


The Daily Threat Landscape refresh on the Databricks Lakehouse — end-to-end architecture

This pipeline runs every morning on its own — raw security data is ingested, deduplicated, correlated, and published as one governed report, with no manual steps in between.



Unity Catalog — one governance layer across every table, the Genie space, the dashboard, and the app's resource bindings.

Lakeflow Job — re-runs the whole Bronze → Silver → Gold refresh automatically every morning (daily cadence).

READS AT A GLANCE

- Raw source tables land as-is (Bronze)
- Normalized + deduplicated (Silver)
- Correlated + aggregated for the briefing (Gold)
- Consumed by dashboard, Genie, and the app (governed)

WHAT CHANGES

BEFORE Fragmented exports pulled separately from five point tools, full of duplicates, with no cross-tool correlation.

AFTER One governed briefing — deduplicated, correlated, and refreshed automatically every morning.

Bronze to Silver — table lineage

How 11 raw Bronze source tables reduce to 3 governed Silver facts through normalization and deduplication.

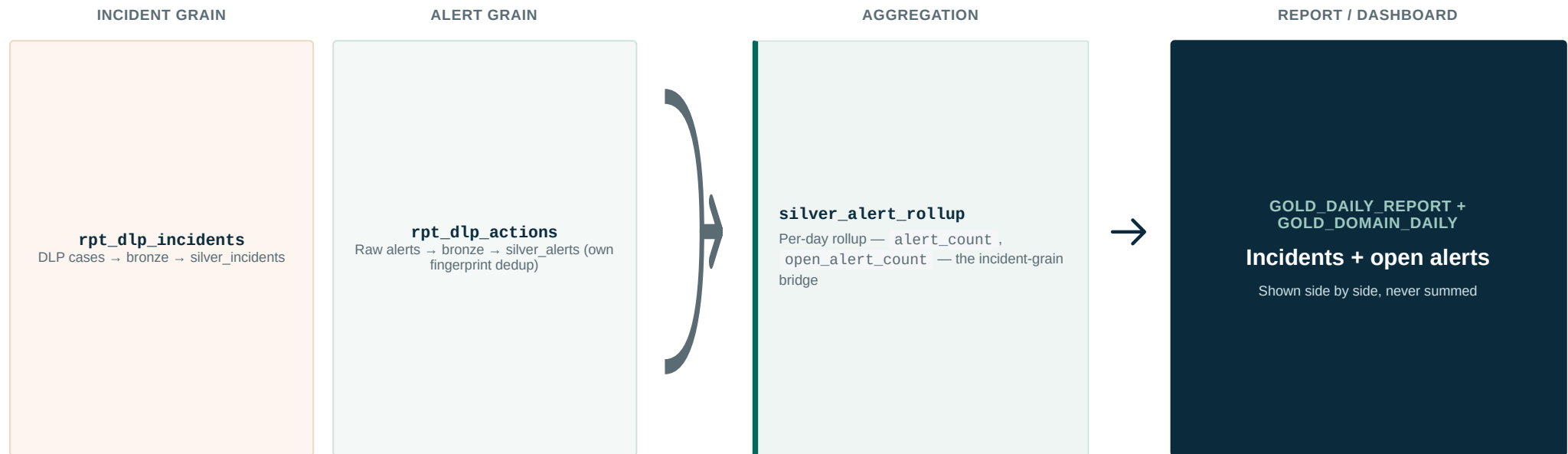


Handling DLP's two grains — incidents vs. alerts

DLP is alert-driven: thousands of raw signals that do not fit the incident schema. Here is how we model it.

The problem — the other four domains are incident-driven (one row = one case a human triages). DLP emits raw **alerts** — machine signals with no case lifecycle, ~10x the volume, and columns (`rule` , `sensor` , `action_taken`) that don't fit the incident schema. Forcing them into the unified table would explode the counts and null out most columns.

THE DECISION — TWO FACTS, ONE BRIDGE (DO NOT UNION ALERTS INTO INCIDENTS)



WHY THIS WINS

- ✓ Incident KPIs stay honest — unique incident counts never inflated by alert volume
- ✓ Alerts still visible as "alert pressure" (`open_alert_count`) on the same report
- ✓ Correlation can cite an alert storm on an affected asset as evidence for the cluster
- ✓ Each grain deduped with its own key
- ✓ Mirrors the real source schema: `rpt_dlp_incidents` (cases) is the parent of `rpt_dlp_actions` (alerts) — the two-grain split is native to DLP

ALTERNATIVES REJECTED

- ✗ Union alerts into `silver_incidents`: counts explode, columns go sparse
- ✗ Fully separate silo, no bridge: report can't show alert pressure, correlation loses the alert signal
- = **Two facts + rollup bridge is the standard medallion multi-grain pattern**

Table glossary — Silver & Gold

What each governed table holds, in one line. Silver normalizes and dedupes; Gold correlates and aggregates for the briefing.

SILVER NORMALIZED, DEDUPLICATED FACTS	
silver_incidents	One unified, deduplicated incident row per case, normalized from the 5 primary domains (CDC/DBR/DLP/Insider/EASM) and enriched with business unit + CTI-bulletin flags.
silver_alerts	DLP raw alerts at alert grain, normalized and fingerprint-deduped (kept separate from incidents).
silver_alert_rollup	Per-day, per-domain rollup of DLP alerts (raw/unique/open counts) that bridges the alert grain back to incident grain.
GOLD CORRELATED, AGGREGATED FOR THE BRIEFING	
gold_daily_report	One row per report date: the daily briefing headline (incident/dup/severity/domain counts, review score, DefCon, cluster count, alert-pressure totals).
gold_domain_daily	Per-date, per-domain rollup: raw vs. unique counts, dedup breakdown, severity/status counts, plus DLP alert columns.
gold_incidents	The denormalized, deduplicated per-incident fact (severity, domain, country, staff, asset + cluster tags) that the dashboard and Genie read row-level.
gold_correlated_clusters	The validated cross-domain attack chains (score ≥ 30), each with a multi-stage narrative connecting the shared entities across correlated incidents and alerts.

Silver descriptions are drawn from the bronze source schema provided. Gold descriptions reflect each table's structural role in the pipeline.

Deploying build_silver_gold.py — from Bronze to governed Silver & Gold

This notebook reads the Bronze tables you already have and normalizes, deduplicates, correlates, and aggregates them into the Silver and Gold layer described in this guide. It does not generate any data.

PREREQUISITES

- ✓ A Unity Catalog catalog + schema you can write to (Bronze may live in the same or a different schema)
- ✓ Your Bronze tables already loaded, matching the expected schema
- ✓ A running cluster (any DBR) or Serverless compute — no extra libraries, pure Spark SQL
- ✓ Permission to CREATE TABLE in the target schema

IMPORT THE NOTEBOOK

OPTION A — WORKSPACE IMPORT (SIMPLEST)

In Databricks: Workspace → your folder → Import. Upload `build_silver_gold.py` — Databricks recognizes it as a Python notebook.

OPTION B — GIT FOLDER / REPO

Commit `build_silver_gold.py` to your repo and clone it as a Databricks Git folder.

CONFIGURE & RUN

WIDGET	WHAT TO SET
Catalog	The Unity Catalog catalog where Silver/Gold are written
Schema	The schema for the output tables (and, by default, where Bronze is read from)

If your Bronze table names differ from the defaults, edit the `B_*` constants in the first code cell (e.g. `B_CDC = "my_cdc_bronze"`). If Bronze lives in a different schema than your output, fully-qualify them (e.g. `B_CDC = "raw_catalog.raw_schema.cdc"`). Then **Run All** — it builds the tables in order and prints row counts + a validation preview. Typical runtime: a couple of minutes.

Expected Bronze tables

Primary incident feeds the notebook reads directly — column names should match, or update the B_* constants.

DOMAIN	DEFAULT BRONZE TABLE	KEY COLUMNS USED
CDC	bronze_snow_latest_incident	number, short_description, u_event_severity, state, category, u_impacted_country, u_reported_by, u_host_name, u_master_ticket, sys_created_on, closed_at, dedup_group, ingestion_date
DBR	bronze_rpt_intr_incidents	id, short_description, severity, state, request_category, impacted_country, assigned_to, created_on, closed_at, dedup_group, ingestion_date
DLP cases	bronze_rpt_dlp_incidents	id, short_description, severity, status, incident_trigger, location_country, sender, machine, correlation_id, created_at, closed_at, dedup_group, ingestion_date
Insider	bronze_snow_latest_record_parsed	INCIDENTID, short_description, u_view_as_intial_business_impact_risk_assesment, state, u_staff_email, u_staff_country, u_cdc_ticket_no_after_advising, sys_created_on, closed_at, dedup_group, ingestion_date
EASM	bronze_watchtowr_findings	id, title, severity, asset, ip, references, status, created_at, dedup_group, ingestion_date

OPTIONAL FEEDS

- ✓ **Alert grain** — bronze_rpt_dlp_actions : id, dlp_incident_id, rule_name, sensor, severity, sender, machine, recipient_destination_url, destination_path, source_file, matches, data_size_mb, action_taken, status, location_country, created_at, ingestion_date
- ✓ **Enrichment** — bronze_rpt_ctpi_incidents (CTI bulletins → has_cti_bulletin); bronze_dbr_business_impacted (→ business_unit)

Optional tables are skipped automatically if not present — the rest still builds.

DEDUP_GROUP — HOW DEDUP IS KEYED

Dedup keeps one canonical row per ingestion group. If your Bronze carries a dedup_group column (grouping an incident with its re-filed duplicates), leave **USE_DEDUP_GROUP = True**. If it doesn't, set **USE_DEDUP_GROUP = False** and the notebook falls back to a content fingerprint (domain, normalized title, country, hour) — review that heuristic against your data.

What it produces — and what to tune

The same governed tables described earlier in this guide, plus two knobs worth reviewing against your data.

LAYER	TABLE	DESCRIPTION
SILVER	silver_incidents	5 domains normalized to one schema, deduplicated, enriched (business_unit, has_cti_bulletin)
SILVER	silver_alerts	DLP raw alerts (alert grain), normalized + fingerprint-deduped
SILVER	silver_alert_rollup	Per-day DLP alert rollup — bridges alert grain to incident grain
GOLD	gold_incidents	Denormalized unique-incident fact + cluster tags — the row-level source
GOLD	gold_correlated_clusters	Validated cross-domain clusters (derived from your data)
GOLD	gold_domain_daily	Per (date, domain) rollup: dedup breakdown, severity, alert pressure
GOLD	gold_daily_report	One row per report date — the briefing headline record

CORRELATION (cell 4)

A cluster = incidents from ≥ 2 domains sharing an entity within a time window.

- ▶ CORRELATION_SIGNALS — which shared entities count. Default ["key"] uses only the explicit cross-tool correlation_key (highest precision). Add "user" and/or "asset" for shared staff email / asset — more coverage, noisier if those recur coincidentally.
- ▶ WINDOW_DAYS (default 14) — shared-entity incidents must fall within this many days.
- ▶ SCORE_THRESHOLD (default 30) — minimum score to be "validated".

REVIEW SCORE (cell 7)

review_score (0–5) is a transparent data-quality heuristic: 5.0 minus the duplicate ratio, so a cleaner day scores higher. Adjust the formula to your own quality definition — for example, adding penalties for missing severity or country fields.

Running it daily & troubleshooting

Once validated, schedule the refresh and connect the consumption layer.

1 RUN IT DAILY (OPTIONAL)

To refresh every morning: Workflows → Create Job → Notebook task → point at this notebook, set the catalog / schema job parameters, and add a daily schedule (e.g. 06:00). Serverless job compute is fine. Each run is idempotent (`CREATE OR REPLACE`).

2 CONNECT THE CONSUMPTION LAYER

Once the Gold tables exist, point at `{catalog}. {schema}` :

- **AI/BI Dashboard** — reads `gold_daily_report`, `gold_domain_daily`, `gold_incidents`, `gold_correlated_clusters`, `silver_alert_rollup`
- **Genie space** — attach the same Gold tables + `silver_alert_rollup`
- **App** — set its catalog/schema config to match

TROUBLESHOOTING

SYMPTOM	FIX
<code>TABLE_OR_VIEW_NOT_FOUND</code> on a Bronze table	Table name/schema mismatch — set the <code>B_*</code> constant (fully-qualify if Bronze is in another schema)
A column can't be resolved	Your Bronze column name differs — adjust that source's <code>SELECT</code> in the Silver <code>UNION</code> (cell 2)
0 correlated clusters	Expected if no cross-tool <code>correlation_key</code> links exist. Add <code>"user"/"asset"</code> to <code>CORRELATION_SIGNALS</code> or widen <code>WINDOW_DAYS</code>
Too many clusters	Narrow <code>CORRELATION_SIGNALS</code> to <code>["key"]</code> , lower <code>WINDOW_DAYS</code> , or raise <code>SCORE_THRESHOLD</code>
No <code>silver_alerts</code>	Expected if <code>bronze_rpt_dlp_actions</code> isn't present — an empty rollup is created so downstream still works